



2IA Practice Problems II

Permutations



Practice Problems Chapter 2

1. Consider the following permutations in S_5 :

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 5 & 3 & 4 \end{pmatrix}, \quad \beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 1 & 2 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 2 & 1 & 3 \end{pmatrix}$$

Compute:

- (a) α^{-1}
 - (b) $\alpha\beta^2$
 - (c) $\gamma\beta^{-1}$
 - (d) $\alpha^{-1}\gamma\beta^{-1}$
2. Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$ be permutations in S_4 . In each case **solve the given equation for χ in S_4 .**
- (a) $\chi\tau = \sigma$
 - (b) $\sigma^{-1}\chi = \tau$
 - (c) $\chi\tau\sigma = \varepsilon$
 - (d) $\tau\chi\sigma^{-1} = \sigma$
3. **Write the following permutations in to matrix-type notation.**
- (a) $\sigma = (1 \ 8 \ 7 \ 4)(3 \ 6 \ 7 \ 5 \ 9)$
 - (b) $\tau = (1 \ 3 \ 5 \ 7)(4 \ 1 \ 9)$
 - (c) $\gamma = (7 \ 1 \ 8 \ 9 \ 2 \ 5 \ 6)(3 \ 6)(4 \ 1 \ 9)$
4. **Factor each of the following permutations into disjoint cycles, and into transpositions. In each case, find the parity.**
- (a) $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 3 & 8 & 9 & 5 & 2 & 1 & 6 & 4 & 7 \end{pmatrix}$
 - (b) $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 6 & 4 & 8 & 9 & 3 & 1 & 7 & 5 & 2 \end{pmatrix}$
 - (c) $\gamma = (1 \ 2 \ 3 \ 4 \ 5)(6 \ 7)(1 \ 3 \ 5 \ 7)(1 \ 6 \ 3)$

5. (Test in 2021) Consider the permutation $\sigma \in S_9$, given below

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 5 & 1 & 6 & 8 & 7 & 3 & 9 & 4 & 2 \end{pmatrix}.$$

- (a) **Find the decomposition into disjoint cycles of σ .**
 - (b) What is the **length of the longest cycle**?
 - (c) **Swap any two element in the second row of the matrix notation of σ to obtain a permutation $\beta \in S_9$ such that the decomposition into disjoint cycles of β has a cycle of longer length.**
6. **In each case determine whether the statement is true or false.** If true, provide an argument supporting your answer; if false, provide a counterexample.
- (a) The product of transpositions $(1 \ 2)(1 \ 3) \dots (1 \ r)$ is indeed r -cycle.
 - (b) If α and β are disjoint permutations in S_n such that $\alpha\beta = \varepsilon$, then $\beta = \alpha^{-1}$.
 - (c) If $\alpha, \beta, \gamma \in S_n$ are such that $\alpha\beta = \gamma\alpha$, then $\beta = \gamma$.
 - (d) The product of any two cycles commute.
 - (e) If σ and τ are distinct transpositions in S_n , then their product $\sigma\tau$ has always order 2.
 - (f) If σ and τ are disjoint transpositions in S_n , then their product $\sigma\tau$ has always order 2.
 - (g) If α and β are not disjoint, then $\alpha\beta \neq \beta\alpha$.
7. (Test 2, 2021) Let $\sigma \in S_n$ be a permutation and $(i_1 \ i_2 \dots i_k)$ a k -cycle in S_n . Prove the following equality:

$$\sigma(i_1 \ i_2 \dots i_k)\sigma^{-1} = (\sigma(i_1) \ \sigma(i_2) \dots \sigma(i_k)).$$